

DEFINICIJE 1

Navadna diferencialna enačba 1. reda $\dot{x} = F(x, t)$

Skalarne linearne enačbe 1. reda $\dot{x} = f(t)x + g(t)$

Integralne krivulje $\dot{y}(t) = \vec{f}(y(t))$

Ekstenzivni izrek $\cdot x_r(0) = x_0 \quad \cdot \dot{x} = F(t, x)$

$\cdot F$ Lipschitzova na $\vec{x} \geq L \quad \cdot F \leq M$ na $[t_0 - a, t_0 + a] \times B_0$

Potem rešitev obstaja in je enolična na

$[t_0 - \alpha, t_0 + \alpha] \rightarrow B_0$, kjer $\alpha \leq \min \left\{ \frac{1}{2M}, \frac{1}{L} \right\}$

Prvi integral u za $y' = F(x, y)$, če za $\forall$ rešitev $y(x)$
velja $u(x, y(x)) \equiv C$

Pfaffova enačba $P(x, y)dx + Q(x, y)dy = 0$

Integrirajoči množitelj $\lambda(x, y)$, da je $\lambda \cdot F$ potencialno

Lagrange d'Alembertova formula $y = x \cdot \varphi(y') + \psi(y')$

Clairnotova enačba $y = x y' + \psi(y')$

Ovojnica družine krivulj $F(x, y; c) \equiv 0$ je $c \mapsto (x(c), y(c))$
za katero velja $\forall c. (x'(c), y'(c)) \perp (\dot{x}(c, t), \dot{y}(c, t))$

DEFINICIJE 2

Picardova preslikava $\mathcal{A}(\vec{x}) = \vec{x}_0 + \int_0^t F(\tau, x(\tau)) d\tau$

Peznov izrek • $F(t, \vec{x}) : [t_0 - a, t_0 + a] \times B_{b, x_0} \longrightarrow \mathbb{R}^n$ zvezna

Potem $\exists x$ da obstaja vsaj ena rešitev $\vec{y} = F(t, \vec{y})$, $\vec{y}(t_0) = \vec{x}_0$

Tok vektorskega polja $\Phi_t(\vec{x}_0) = \vec{y}(t, x_0)$ za $\vec{y}$ rešitev problema $\dot{\vec{y}} = F(t, \vec{y})$ $y(t_0) = x_0$

Infinitesimalni generator $\{\Phi_t\}$ je polje, da velja

$$\left. \frac{d}{dt} \right|_{t=0} \Phi_t(x) = F(x)$$

Kompletno polje F , če za $\forall x_0 \in \Omega$ je rešitev problema $\dot{\vec{x}} = F(\vec{x})$, $\vec{x}(0) = \vec{x}_0$ definirana za $\forall t \in \mathbb{R}$

Hamiltonski sistem $\Omega \subseteq \mathbb{R}^{2n}$ $\Omega = \{(q_1, \dots, q_n, p_1, \dots, p_n)\}$

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

Liouvillova enačba • Φ_t tok vekt. polja F na $\Omega \subseteq \mathbb{R}^{2n}$

$$\frac{d}{dt} \det(J\Phi_t(\vec{a})) = \text{div}(F(\Phi_t(\vec{a}))) \cdot \det(J\Phi_t(\vec{a}))$$

$$\text{Oziram } \det(J\Phi(\vec{a})) = e^{\int \text{div}(F(\tau, \vec{a})) d\tau}$$

Hamiltonsko polje $X_H = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \nabla H$

Nepopolni eliptični integral 1. vrste $F(z, k) = \int_0^z \frac{d\tau}{\sqrt{1 - k^2 \sin^2(\tau)}}$

Jacobijeva amplituda $\text{am}(z, k) = F^{-1}(z, k) \nearrow$

Poissonov oklepaj $\{f, g\} = \sum_{i=1}^n \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right) = X_f \cdot \nabla g$

Liejev produkt vektorskih polj

• antisimetričnost $[F, G] = -[G, F]$

• Jacobijeva identiteta

$$[E, [F, G]] + [G, [E, F]] + [F, [G, E]] = 0$$

Simetrija Hamiltonskega sistema (H, Ω) je difeomorfizem

$S: \Omega \longrightarrow \Omega$, da γ rešitev za (H, Ω) $\gamma(t) = (p(t), q(t))$

$\gamma(0) = x_0$, potem je $S(\gamma(t))$ rešitev za (H, Ω) z

začetnim pogojem $S(\gamma(0)) = S(x_0)$

DEFINICIJE 3

$$\dot{\vec{x}} = A(t) \vec{x}$$

Fundamentalna matrika $\Phi_{t_0, t}$ sistema $\dot{\vec{x}} = A(t) \vec{x}$ je
 $\Phi = [\vec{x}_1 \dots \vec{x}_n]$ kjer so $\vec{x}_i$ baza rešitev. $\vec{x}_i(t_0) = \vec{e}_i$

Ekspencijacija matrike $e^A = I + \sum_{n=1}^{\infty} \frac{1}{n!} A^n$

BCH formula $e^A e^B = e^C \leadsto C = \ln(e^A e^B)$

Dysonova vrsta $\Phi_t = I + \sum_{n=1}^{\infty} \int_{\mathcal{D}_n(t)} A(\tau_1) A(\tau_2) \dots A(\tau_n) d\vec{\tau}$

Liouvillova formula $\det(\Phi_t) = \det(\Phi_{t_0}) \exp \int_{t_0}^t \text{tr}(A(\tau)) d\tau$
 $\det(\Phi_t) = \det(\Phi_{t_0}) e^{\int_{t_0}^t \text{tr}(A(\tau)) d\tau}$

Determinanta wronskega $W(t) = \det(\Phi_t)$

DEFINICIJE 4

Lagrangeev funkcional $\mathcal{L}(y) = \int_a^b L(x, y(x), y'(x)) dx$

Šibki odvod (smerni, Gâteauxov) $P: U \rightarrow V$

$$U_{u_0}^{\circ}(v) = \lim_{t \rightarrow 0} \frac{1}{t}(P(u_0 + tv) - P(u_0)) = \left. \frac{d}{dt} \right|_{t=0} P(u_0 + tv)$$

Krepi odvod (Frechetov) $P: U \rightarrow V$, $A: U \rightarrow V$ linearna, omejena. $P(u_0 + v) = P(u_0) + A(v) + o(u_0, v)$; $\lim_{\|v\| \rightarrow 0} \frac{\|o\|}{\|v\|} = 0$

Dopustna funkcija $v \in \{u \in U, u(a) = A, u(b) = B\}$

Dopustna variacija $v \in \{u \in U, u(a) = 0, u(b) = 0\}$

Euler-Lagrangeova enačba $\frac{\partial L}{\partial y}(x, y(x), y'(x)) - \frac{d}{dx} \frac{\partial L}{\partial y'}(\dots) = 0$

Vezani ekstrem: Pogoji $\Phi(u) = l \in \mathbb{R}$, ekstrem $\mathcal{L}$

KAJ MORAM ZNATI

- Kako vemo, da je rešitev sistema globalno definirana
- Dysonova vrsta
- Liouvillova formula
- Vezen: ekstremi
- Euler-Lagrangova enačba
- Tangentni prostor je enake jedru vezi
- Eksistenčni izrek
- Variacijski račun
- odvod L
- dopustne variacije
- tok in lastnosti
- primeri za eksistenčni izrek
- Fundamentalne metrike

Eksistenčni izrek

• $F(t, \vec{x}) : [t_0 - a, t_0 + a] \times B_b \rightarrow \mathbb{R}$ zvezna, Lipschitzova na drugi spremenljivki s konstanto L .

• $\dot{\vec{y}} = F(t, \vec{y}) \quad \vec{y}(t_0) = \vec{x}_0 \in B_{\frac{b}{2}}$

Potem obstaja natanko ena rešitev problema

$$\vec{y} : [t_0 - \alpha, t_0 + \alpha] \rightarrow B_b \quad \alpha \leq \min \left\{ \frac{b}{2M}, \frac{1}{L} \right\}$$

$$M = \max \{ F(t, \vec{x}) ; (t, \vec{x}) \in [t_0 - a, t_0 + a] \times B_b \}$$

Dokaz:

$$\vec{x}(t) = \vec{x}_0 + \int_{t_0}^t F(\tau, \vec{x}(\tau)) d\tau$$

$$\mathcal{A}(\vec{x}) = \vec{x}_0 + \int_{t_0}^t F(\tau, \vec{x}(\tau)) d\tau \quad \mathcal{A} : V \rightarrow V$$

$\vec{x}$ je rešitev $\Leftrightarrow \vec{x}(t) = \mathcal{A}(\vec{x}(t)) \Leftrightarrow \vec{x}$ je fiksna točka.

Take točke obstaja, ko je $\mathcal{A}$ skrajitev

$$P_{a,b} = \{ (t, \vec{x}) \in \mathbb{R} \times \mathbb{R}^n ; |t - t_0| < a, \|\vec{x} - \vec{x}_0\| \leq b \}$$

$$E = \{ \vec{y} : \mathcal{I} \rightarrow \mathbb{R}^n ; \vec{y}(t_0, \vec{x}) = \vec{x} ; \|\vec{y}(t, \vec{x}) - \vec{x}_0\| \leq b \quad \forall t, \vec{x} \in P_{a,b/2} \}$$

$$\mathcal{I} = [t_0 - \alpha, t_0 + \alpha] \times B_{\frac{b}{2}}$$

$$\mathcal{A}(E) \subseteq E$$

$$1) \mathcal{A}(\vec{y}(t_0, \vec{x})) = \vec{x} \quad \mathcal{A}(\vec{y}) - \vec{x} = \int_{t_0}^t F(\tau, \vec{y}(\tau, \vec{x})) d\tau = 0$$

$$2) \|\mathcal{A}(\vec{y}(t, \vec{x})) - \vec{x}_0\|_\infty \leq \|\vec{x} - \vec{x}_0\| + \int_{t_0}^t \|F(\tau, \vec{y}(\tau, \vec{x}))\| d\tau \leq \frac{b}{2} + \alpha M \leq b \quad \text{za} \quad \alpha \leq \frac{b}{2M}$$

Ali je $\mathcal{A}$ na E skrajitev?

$$\begin{aligned} \|\mathcal{A}(\vec{y}_1) - \mathcal{A}(\vec{y}_2)\| (t, \vec{x}) &= \left\| \int_{t_0}^t F(\tau, \vec{y}_1) - F(\tau, \vec{y}_2) d\tau \right\|_\infty \leq \\ &\leq \left\| \int_{t_0}^t L \|\vec{y}_1(\tau, \vec{x}) - \vec{y}_2(\tau, \vec{x})\| d\tau \right\|_\infty \leq \alpha L \|\vec{y}_1 - \vec{y}_2\|_\infty \\ \alpha L < 1 &\Rightarrow \alpha < \frac{1}{L} \end{aligned}$$

Torej rešitev obstaja

Zvezna odvisnost od začetne točke

$$y_1(t, x_1) \quad y_2(t, x_2) ; \|x_1 - x_2\| < \delta \Rightarrow y_1(t, x_1) - y_2(t, x_2)$$

$$\|\mathcal{A}(t, y_1(t, x_1)) - \mathcal{A}(t, y_2(t, x_2))\|_\infty \leq$$

$$\leq \|x_1 - x_2\| + \left\| \int_{t_0}^t L \|y_1(t, x_1) - y_2(t, x_2)\| dt \right\|_\infty \leq$$

$$\leq \delta + \left\| \int_{t_0}^t L \|y_1 - y_2\|_\infty dt \right\|_\infty = \delta + L \|y_1 - y_2\|_\infty \cdot \alpha$$

$$\|y_1 - y_2\|_\infty (1 - L\alpha) \leq \delta$$

$$\|y_1 - y_2\|_\infty \leq \frac{\delta}{1 - L\alpha}$$

Tok vektorskega polja

$$\dot{\Phi}_t(x) = F(t, \Phi_t) \quad / \quad \frac{d}{dx}$$

$$(D_x \dot{\Phi}_t)(h) = (D_x F)(t, \Phi_t) \cdot (D_x \Phi_t)(h)$$

$$\leadsto \frac{d}{dt} (D_x \Phi) = D_x F(t, \Phi_t) \cdot D_x \Phi_{t_0} \quad D_x \Phi_{t_0} = I$$

Pomembni lastnosti:

$$1) \Phi_{t_0, t_0}(\vec{x}) = \vec{x}$$

$$2) \Phi_{t, s} \circ \Phi_{s, t_0} = \Phi_{t, t_0}$$

$$\text{Autonomni sistem} \leadsto \Phi_{t, t_0} = \Phi_{t+\tau, t_0+\tau} \Rightarrow \Phi_{t, t_0} = \Phi_{s, 0}$$

$$\leadsto 1) \Phi_0 = I$$

$$2) \Phi_{s+t} = \Phi_s \circ \Phi_t$$

$$\Rightarrow \Phi_t^{-1} = \Phi_{-t} \Rightarrow \Phi_t \text{ je difeomorfizem}$$

Izrek: Liouvilleova enačba

Φ_t tok avtonomnega polja $F(\vec{x})$ $J\Phi = \det(D\Phi)$

Potem $\frac{d}{dt} d\Phi_t(\vec{x}) = \operatorname{div}(F(\Phi_t(\vec{x}))) d\Phi_t(\vec{x})$

Dokaz: (za $n=2$)

Oznacimo $\Phi_t \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a(t, x, y) \\ b(t, x, y) \end{bmatrix}$

$$\dot{\Phi}_t \begin{bmatrix} x \\ y \end{bmatrix} = F(\Phi_t \begin{bmatrix} x \\ y \end{bmatrix}) \leadsto \begin{bmatrix} \dot{a} \\ \dot{b} \end{bmatrix} = F(\begin{bmatrix} a \\ b \end{bmatrix}) / D_x$$

$$D_{\vec{x}}(\partial_t \begin{bmatrix} a \\ b \end{bmatrix}) = D_{\vec{x}} \left(\begin{bmatrix} f(a, b) \\ g(a, b) \end{bmatrix} \right)$$

$$\parallel$$

$$(D_{\vec{x}} \begin{bmatrix} a \\ b \end{bmatrix}) = \begin{bmatrix} \partial_x a & \partial_y a \\ \partial_x b & \partial_y b \end{bmatrix}$$

$$\begin{aligned} J &= \partial_t \det(D_x(\Phi_t \begin{bmatrix} x \\ y \end{bmatrix})) = \partial_t \det \begin{bmatrix} \partial_x a & \partial_y a \\ \partial_x b & \partial_y b \end{bmatrix} = \\ &= \partial_t (\partial_x a \cdot \partial_y b - \partial_y a \cdot \partial_x b) = \\ &= \partial_t \partial_x a \cdot \partial_y b + \partial_x a \partial_t \partial_y b - (\partial_t \partial_y a \cdot \partial_x b + \partial_y a \cdot \partial_t \partial_x b) = \\ &= \partial_x \partial_t a \cdot \partial_y b - \partial_y \partial_t a \cdot \partial_x b + \partial_x a \cdot \partial_y \partial_t b - \partial_y a \cdot \partial_x \partial_t b = \\ &= \begin{vmatrix} \partial_x \dot{a} & \partial_y \dot{a} \\ \partial_x b & \partial_y b \end{vmatrix} + \begin{vmatrix} \partial_x a & \partial_y a \\ \partial_x \dot{b} & \partial_y \dot{b} \end{vmatrix} \end{aligned}$$

$$\text{ker velja } \begin{bmatrix} \dot{a} \\ \dot{b} \end{bmatrix} = F(\begin{bmatrix} a \\ b \end{bmatrix}) = \begin{bmatrix} f(a, b) \\ g(a, b) \end{bmatrix}$$

$$\text{je } \dot{a} = f(a, b) \leadsto \partial_x \dot{a} = f_a \partial_x a + f_b \partial_x b$$

$$\begin{aligned} J &= \begin{vmatrix} f_a \partial_x a + f_b \partial_x b & f_a \partial_y a + f_b \partial_y b \\ \partial_x b & \partial_y b \end{vmatrix} + \\ &+ \begin{vmatrix} \partial_x a & \partial_y a \\ g_a \partial_x a + g_b \partial_x b & g_a \partial_y a + g_b \partial_y b \end{vmatrix} = \end{aligned}$$

$$\begin{aligned} &= f_a (\partial_x a \cdot \partial_y b - \partial_y a \partial_x b) + f_b (\partial_x b \partial_y b - \partial_y b \partial_x b) \\ &+ g_a (\partial_x a \partial_y a - \partial_y a \partial_x a) + g_b (\partial_x a \partial_y b - \partial_y a \partial_x b) = \\ &= \begin{bmatrix} f_a \\ g_b \end{bmatrix} \cdot \begin{vmatrix} \partial_x a & \partial_y a \\ \partial_x b & \partial_y b \end{vmatrix} = \operatorname{div} F(\Phi_t \begin{bmatrix} x \\ y \end{bmatrix}) \cdot \det(D_x \Phi_t \begin{bmatrix} x \\ y \end{bmatrix}) \end{aligned}$$